

STABIO, Switzerland

WMO: 067710

Lat: 45.8435N

Lon: 8.9324E

Elev: 1156

StdP: 14.09

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.4	(c) 21.5	(d) 1.4	(e) 6.2	(f) 37.4	(g) 6.4	(h) 8.0	(i) 39.2	(j) 12.0	(k) 47.7	(l) 9.8	(m) 46.4	(n) 2.1	(o) 230	(p) 0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.8	(c) 87.6	(d) 70.6	(e) 85.0	(f) 69.4	(g) 82.7	(h) 68.1	(i) 72.8	(j) 83.9	(k) 71.4	(l) 81.7	(m) 70.0	(n) 79.6	(o) 4.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 69.4	(b) 112.8	(c) 77.5	(d) 68.0	(e) 107.4	(f) 76.1	(g) 66.6	(h) 102.4	(i) 74.7	(j) 37.2	(k) 84.0	(l) 35.8	(m) 81.8	(n) 34.7	(o) 79.7	(p) 81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.5	(b) 7.5	(c) 6.1	(e) 14.1	(f) 91.3	(g) 4.4	(h) 2.6	(i) 11.0	(j) 93.2	(k) 8.4	(l) 94.7	(m) 5.9	(n) 96.2	(o) 2.7	(p) 98.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			13.0	75.7	4.3	2.0	9.9	77.1	7.3	78.3	4.9	79.4	1.7	80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.5	36.1	38.4	45.8	52.4	60.2	67.8	71.4	70.3	62.5	54.6	44.8	37.2	(6)
(7)		DBStd	13.53	5.15	5.49	5.85	5.47	4.95	5.28	4.31	4.34	4.94	5.48	5.86	5.48	(7)
(8)		HDD50	1538	433	326	152	37	1	0	0	0	0	21	170	398	(8)
(9)		HDD65	4715	897	746	595	380	161	31	5	8	103	322	605	862	(9)
(10)		CDD50	2830	1	0	22	108	319	534	662	628	376	164	15	1	(10)
(11)		CDD65	531	0	0	0	0	14	114	202	171	29	1	0	0	(11)
(12)		CDH74	4516	0	0	5	20	184	1056	1740	1297	202	12	0	0	(12)
(13)		CDH80	1181	0	0	0	2	20	281	508	350	19	1	0	0	(13)
(14)	Wind	WSAvg	2.1	1.9	2.0	2.4	2.5	2.3	2.3	2.2	1.9	1.9	1.8	1.8	1.8	(14)
(15)	Precipitation	PrecAvg	62.6	3.0	2.8	3.1	5.5	7.1	6.0	5.7	6.7	6.4	7.6	3.0		(15)
(16)		PrecMax	93.7	9.1	11.2	8.9	12.6	19.6	16.4	12.0	12.4	17.6	17.0	26.8	9.1	(16)
(17)		PrecMin	38.3	0.1	0.1	0.2	0.5	1.7	0.7	1.8	0.6	0.6	0.4	0.0	0.0	(17)
(18)		PrecStd	14.5	2.5	2.6	2.2	3.3	3.9	3.3	2.9	3.0	4.3	4.0	6.5	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.7	63.0	73.0	76.4	82.6	89.5	90.9	90.8	82.4	74.7	63.8	57.6	(19)
(20)			MCWB	46.7	46.7	52.2	59.0	64.7	70.8	72.0	72.4	67.4	60.6	52.1	42.8	(20)
(21)		2%	DB	52.7	57.0	66.9	72.4	78.8	86.3	87.7	86.8	79.3	70.3	59.7	52.4	(21)
(22)			MCWB	42.0	44.4	50.5	56.5	62.8	69.7	70.5	71.0	66.4	60.1	51.0	42.8	(22)
(23)		5%	DB	49.0	53.3	63.1	69.2	75.5	83.4	85.2	83.8	76.1	67.1	56.5	49.5	(23)
(24)			MCWB	40.0	42.5	49.2	54.7	61.2	68.1	69.5	69.3	63.9	58.4	50.2	42.3	(24)
(25)		10%	DB	45.8	49.7	59.2	65.4	72.4	80.3	82.8	81.1	73.0	63.9	54.1	46.6	(25)
(26)			MCWB	39.1	40.3	47.2	52.8	59.6	66.6	68.4	68.0	62.3	57.3	49.2	41.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	48.8	54.5	60.5	67.3	74.2	74.8	74.4	70.1	64.2	56.4	49.0	(27)
(28)			MCDWB	55.8	58.9	67.8	73.6	78.6	85.3	85.5	85.4	79.0	71.6	60.1	51.1	(28)
(29)		2%	WB	44.6	46.3	52.4	58.2	64.8	71.6	72.9	72.6	68.0	62.3	54.0	46.3	(29)
(30)			MCDWB	49.7	53.8	63.3	69.5	75.6	82.1	84.0	83.5	76.4	67.2	56.9	49.0	(30)
(31)		5%	WB	42.5	44.2	50.8	56.2	63.0	69.8	71.4	71.1	66.3	60.7	52.4	44.4	(31)
(32)			MCDWB	46.6	50.4	59.6	66.0	72.9	80.4	81.7	81.2	73.0	65.0	54.8	47.0	(32)
(33)		10%	WB	40.4	42.2	49.0	54.4	61.2	68.1	70.0	69.8	64.6	58.9	50.6	42.5	(33)
(34)			MCDWB	44.0	47.1	56.7	62.7	69.9	77.9	79.6	78.6	70.4	62.4	52.8	45.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.2	19.4	21.8	20.5	19.6	20.3	20.8	19.8	18.5	16.3	15.4	16.3	(35)
(36)			MCDBR	23.0	27.2	30.4	27.8	25.7	24.5	23.7	23.5	22.6	21.2	20.0	21.3	(36)
(37)			MCWBR	14.6	16.7	16.7	14.1	12.3	10.7	10.0	10.4	11.0	12.0	12.5	13.8	(37)
(38)		5% WB	MCDWR	19.3	23.5	26.2	23.8	22.6	21.9	21.8	21.4	19.2	17.1	14.4	15.7	(38)
(39)			MCWBR	13.3	15.5	15.2	12.6	11.6	10.1	10.0	10.1	10.0	10.3	10.1	11.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.358	0.410	0.460	0.464	0.483	0.464	0.457	0.445	0.422	0.362	0.320	(40)	
(41)		taud	2.339	2.262	2.160	2.062	2.105	2.100	2.161	2.189	2.184	2.198	2.325	2.381	(41)	
(42)		Ebn at Noon	242	255	256	253	257	252	255	252	242	227	224	231	(42)	
(43)		Edh at Noon	24	32	41	49	49	49	46	43	40	34	25	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	477	748	1158	1425	1679	1948	1973	1682	1276	780	469	397	(44)	
(45)		RadStd	65	101	154	184	146	132	137	118	100	112	79	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	+1.48	+0.77	N/A	N/A	N/A	+117	+106
(47) Regional (102 neighbors)		N/A	N/A	N/A	+1.30	+0.72	+0.58	N/A	N/A	+108	+29

Nomenclature: See separate page